

Pain Detection via Knowledge Distillation

A comparative study of Teacher Architectures

1 Introduction

- Non-verbal patients cannot communicate pain (infants, cognitive impairments, unconscious)
- Previous work: 84.15% accuracy (El Morabit & Rivenq, 2022)



Fig. 1: AI-based pain detection system for non-verbal patients.

3 Methodology

Knowledge Distillation Framework

Teachers: ResNet-50 (CNN) vs Swin (Transformer)

Student: DeiT (Vision Transformer)

Knowledge Distillation Architecture for Pain Detection

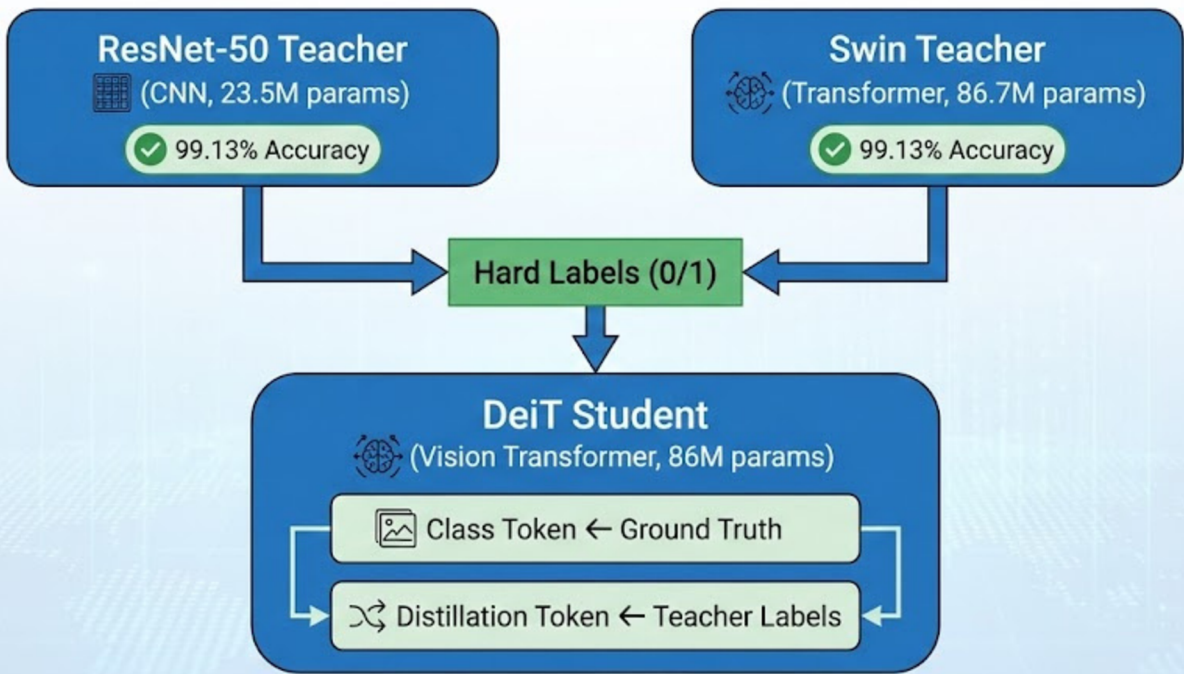


Fig. 3: Knowledge distillation framework. Two teachers (ResNet-50 CNN, Swin Transformer) both fine-tuned to 99.13% accuracy provide hard labels to train DeiT student with dual-token architecture.

4 Experiment Analysis and results

3.1 Datasets

SynPain: AI-generated facial expressions

- Total: 10,710 images
- Classes: NoPain (74.5%), Pain (25.5%)
- Split: 70% train / 15% val / 15% test
- Preprocessing: MTCNN detection → 256×256



Fig. 4: Top: Pain expressions with raised cheeks, squinted eyes. Bottom: Neutral expressions. Diverse age, gender, ethnicity.

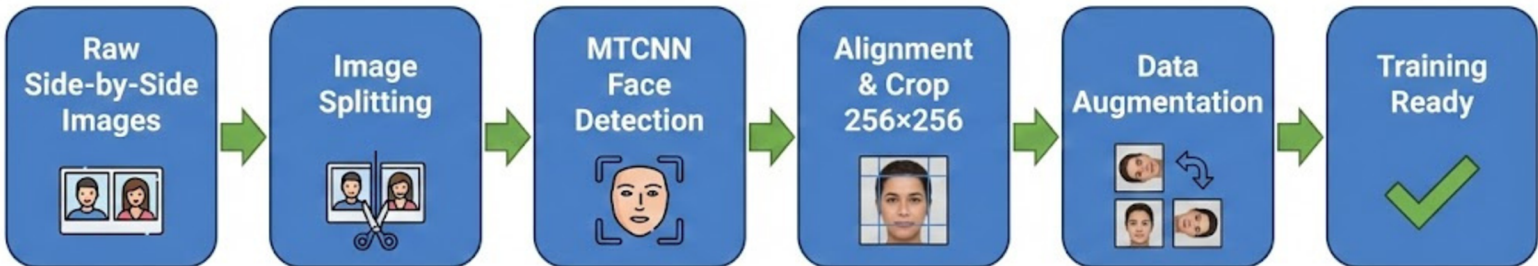


Fig. 5: Data preprocessing pipeline. Side-by-side images split, faces detected via MTCNN, aligned and cropped to 256×256, then augmented for training.

5 Conclusion

✗ Architectural alignment did NOT improve distillation
ResNet50 > Swin
96.45% vs 90.04%
WHY: Teacher Quality > Architectural Alignment

References

S. El Morabit and A. Rivenq, "Pain Detection From Facial Expressions Based on Transformers and Distillation," in 2022 11th International Symposium on Signal, Image, Video and Communications (ISIVC), Valenciennes, France, 2022, pp. 1-5, doi: 10.1109/ISIVC54825.2022.9800746.
TOUVRON, Hugo, CORD, Matthieu, DOUZE, Matthijs, et al. Training data-efficient image transformers & distillation through attention. In : International Conference on Machine Learning. PMLR, 2021. p. 10347 10357.

2 Objectives

RESEARCH QUESTION:

Does architectural alignment between teacher and student improve knowledge distillation efficiency?
Compare ResNet50 vs Swin as teachers for DeiT student in pain detection from facial expressions

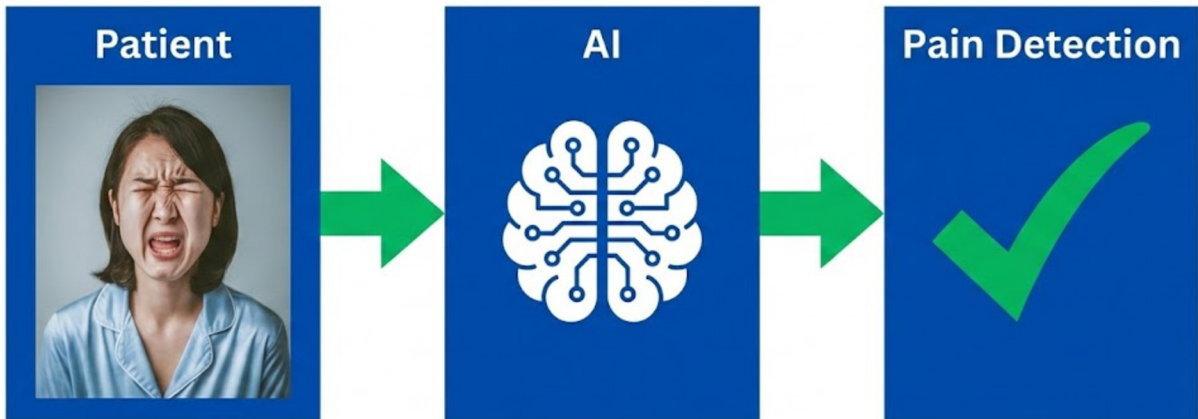


Fig. 2: Model analyzes facial expressions to automatically identify pain presence.

3.2 Results

Metric	ResNet50 Teacher	Swin Teacher
Fine-tune Val Acc	99.13%	99.13%
Test Accuracy	~99%	82.51%
Pain Precision	~94%	100%
Pain Recall	~92%	31.6%
Fine-tune Time	37.4 min	547.5 min

Table 1: Teacher comparison. Equal validation (99.13%) but divergent test performance. Swin's poor recall (31.6%) indicates precision-biased predictions unsuitable for pain detection.

Metric	ResNet50 → DeiT	Swin → DeiT	Winner
Accuracy	96.45%	90.04%	ResNet ✓
Precision (Pain)	94.25%	75.25%	ResNet ✓
Recall (Pain)	91.73%	90.99%	Tie
F1-Score (Pain)	92.97%	82.38%	ResNet ✓
ROC-AUC	0.983	0.955	ResNet ✓

Table 2: Student knowledge distillation results. ResNet50→DeiT achieves 96.45% accuracy, outperforming Swin→DeiT (90.04%) across all metrics, demonstrating teacher quality > architectural alignment.

ResNet50 → DeiT		Swin → DeiT	
NoPain	Pain	NoPain	Pain
1173	23	1073	123
34	377	37	374
98.1% NoPain recall 91.7% Pain recall BALANCED ✓		89.7% NoPain recall 91.0% Pain recall IMBALANCED ✗	

Fig. 6: Confusion matrices. ResNet50→DeiT balanced (98.1%/91.7%). Swin→DeiT imbalanced (89.7%/91.0%).

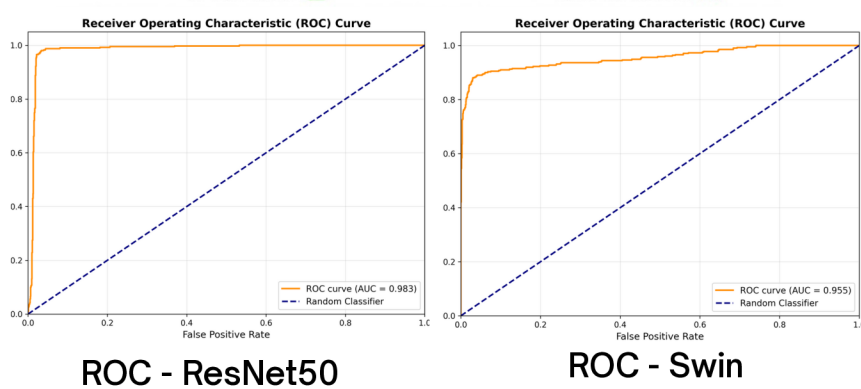


Fig. 7: ROC curves. ResNet50→DeiT superior (AUC=0.983) vs Swin→DeiT (AUC=0.955).

6 Future Work

- Validate on real clinical datasets (UNBC-McMaster, BioVid)
- Investigate soft vs hard distillation approaches
- Explore ensemble teachers (ResNet50 + Swin combined)